



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2013
Higher 1

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H1 BIOLOGY

8875/02

Paper 2

18 September 2013

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination,

1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
 - Section A
 - Section B

The number of marks is given in brackets [] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
Section A	
1	/ 9
2	/ 10
3	/ 10
4	/ 11
Section B	
5 / 6	/ 20
Total	/100

This paper consists of ___ printed pages.

[Turn over]

Section A

Answer **all** the questions in this section.

QUESTION 1

Heavy metal ions are waste products of many industrial processes and at times may leak into the ocean and cause environmental pollution. The toxicity of heavy metals on organisms stems from their effect on metabolically-important enzymes, such as lactate dehydrogenase. Lactate dehydrogenase catalyses the conversion of pyruvate into lactate.

Fig. 1.1 shows the effect of varying concentrations of mercury ions on the rate of reaction catalysed by lactate dehydrogenase isolated from *Oreochromis mossambicus*, a species of fish.

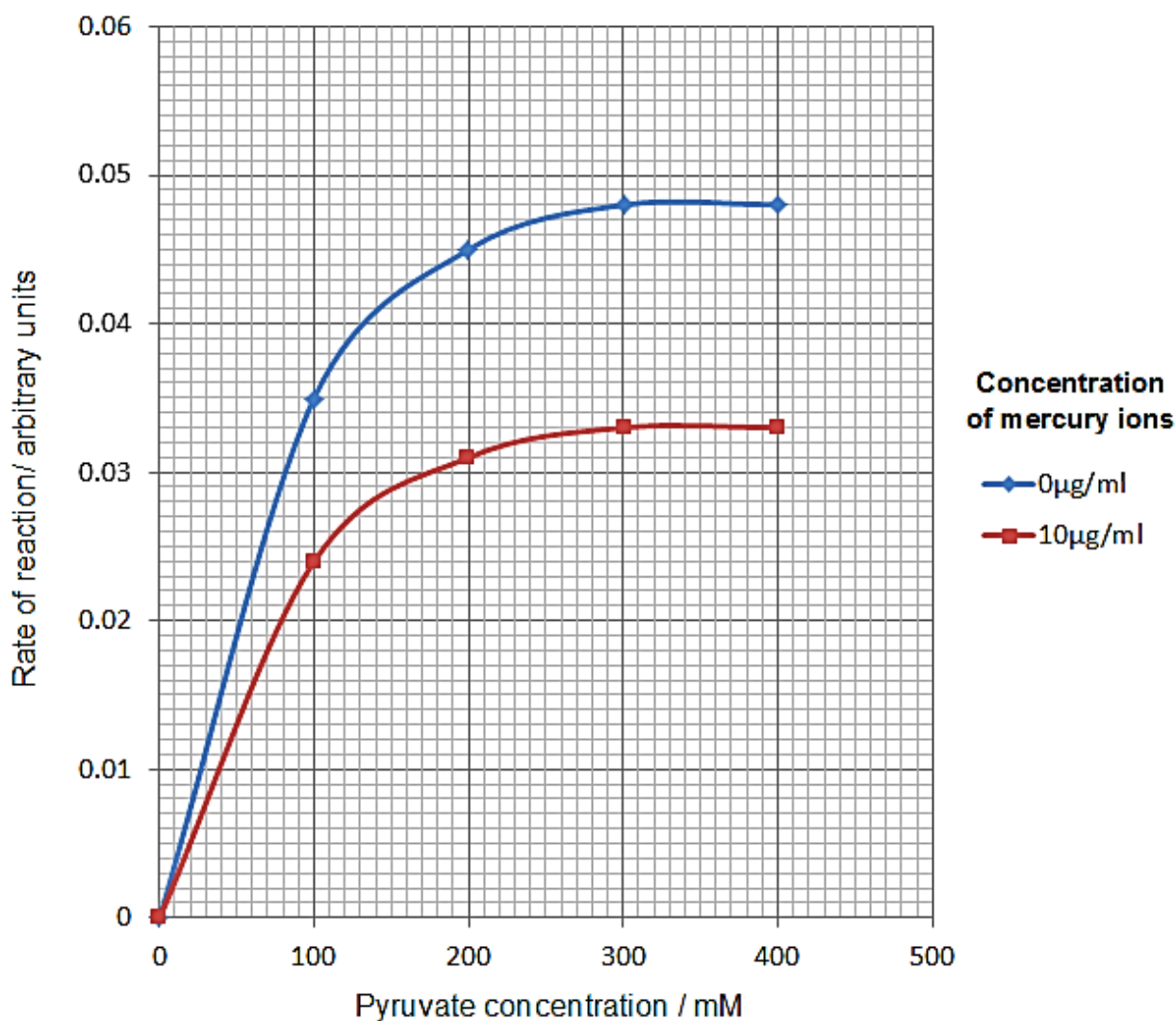


Fig. 1.1

(a) Describe and explain the effect of mercury ions on the rate of reaction catalysed by lactate dehydrogenase. [4]

- In the presence of 10µg/ml of mercury, the maximum rate (V_{max}) of reaction decreased from 0.048 arbitrary units to 0.033 arbitrary units.

Accept: other valid citation

- Mercury ions act as a non-competitive inhibitor...
- Has a different structure from the substrate pyruvate and binds permanently / irreversibly to another site away from the active site on the enzyme
Forms a strong covalent bond to an amino acid R group (e.g. -SH group) on the enzyme.
- This causes a conformational change on the enzyme which alters the shape of the active site.
- Hence the enzyme is rendered non-functional / no longer able to bind to substrate / decreased rate of formation of enzyme-substrate complex and hence decreasing the rate of reaction.
- Increasing substrate/pyruvate concentration will not allow the maximum rate of reaction to be achieved.

(b) Explain how mercury ions may affect the survival of the fish when oxygen concentrations are low. [3]

- When oxygen concentrations are low, aerobic respiration cannot produce sufficient ATP.
- In addition, the fish cannot carry out anaerobic respiration in the presence of mercury ions...
- ... as lactate dehydrogenase cannot catalyse the conversion of pyruvate to lactate/lactic acid to regenerate NAD⁺ in the cytoplasm due to inhibition by mercury ions.
- Hence, glycolysis cannot continue to produce ATP via substrate level phosphorylation.
- Thus ATP-requiring cellular activities are inhibited, causing the death of the fish.

Human lysozyme is an important enzyme which is part of the human body's defence against some bacteria. It is found in abundance in tears, saliva and mucous. The main action of lysozyme is to catalyse the hydrolysis of β -(1,4) glycosidic bonds between N-acetylmuramate (NAM) and N-acetylglucosamine (NAG) sugar residues in the peptidoglycan cell walls of bacteria. With the weakened cell wall, the bacterial cells are more likely to undergo lysis.

Fig. 1.2 shows the structure of lysozyme

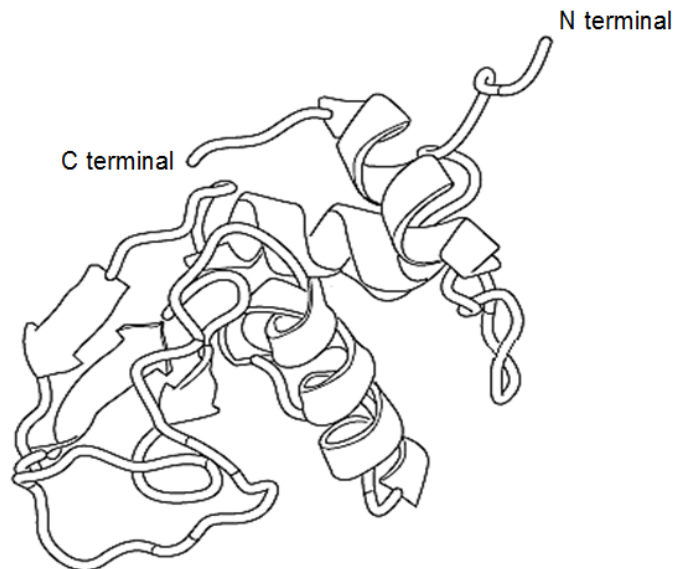


Fig. 1.2

(c) With reference to Fig. 1.2 and with your knowledge on haemoglobin, state two structural differences between lysozyme and haemoglobin. [2]

- The secondary structure of lysozyme consists of α -helices and β -pleated sheets, whereas the secondary structure of haemoglobin consists of only α -helices.
- Lysozyme is a protein with a tertiary structure as it consists of only one polypeptide chain, whereas haemoglobin is a protein with a quaternary structure as it consists of four polypeptide chains.
- Lysozyme does not carry a prosthetic group, whereas each polypeptide chain in haemoglobin carries a prosthetic (non-proteinaceous) haem group containing Fe^{2+} which binds a molecule of oxygen each.

[Total: 9]

QUESTION 2

Fig. 2.1 illustrates the Central Dogma, which describes the flow of genetic information of DNA to RNA to proteins.

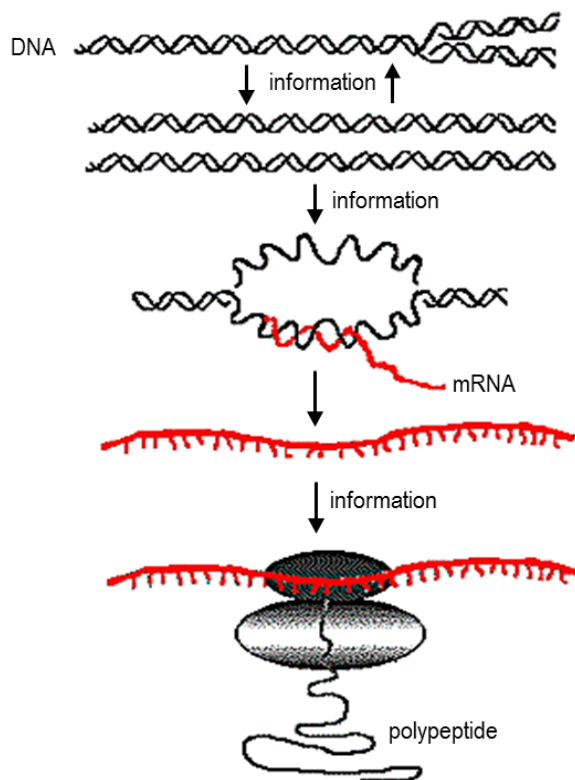


Fig. 2.1

(a) Describe how the information carried on mRNA is related to its role in translation in a prokaryotic cell. [3]

- 5' UTR allows recognition by small ribosomal subunit / translation factors
- Start codon signals the start of translation
- Three bases on the mRNA correspond to a codon
- Each codon on mRNA codes for a specific amino acid
- The stop codon (UAG, UAA, UGA) terminates translation by allowing the binding of a release factor that adds a water molecule to the polypeptide chain

A ribonucleoprotein is a protein that is associated with RNA. The ribosome is an example of a ribonucleoprotein. Prokaryotes have 70S ribosomes while eukaryotes have 80S ribosomes, each consisting of a small and large subunit. In prokaryotes, the small subunit is made up of a 16S rRNA associated with 21 proteins. Fig. 2.2 shows the structure of 16S rRNA.

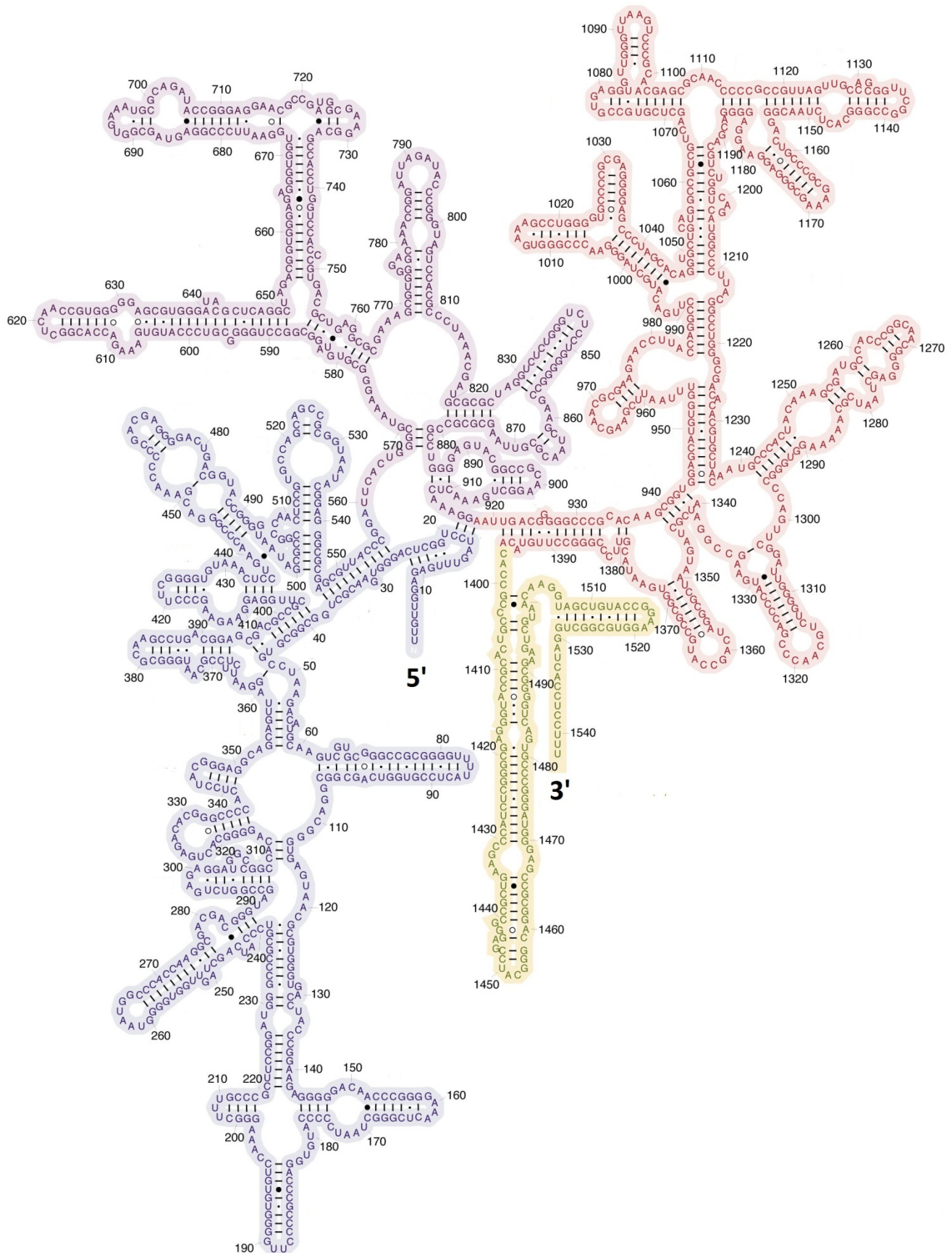


Fig. 2.2

- (b)** 16S rRNA and tRNA are both made up of four types of ribonucleotides linked by phosphodiester bonds.

State two other similarities between the structures of the 16S rRNA and tRNA. [2]

- Both rRNA and tRNA are single stranded molecules.
- Both have double-stranded regions / can form loops
- Both have intra-molecular hydrogen bonds between complementary bases / between A&U and C&G.

- (c)** Describe how eukaryotic ribosomes are formed. [4]

- rRNA is produced through transcription of the rRNA genes in the nucleolus
- Genes of ribosomal proteins are transcribed in nucleus to form mRNA
- which is translated in cytoplasm to form ribosomal proteins
- Ribosomal protein is transported from the cytoplasm into nucleus via nuclear pore
- Partial assembly of rRNA and ribosomal proteins into large and small subunits occurs in nucleolus
- The large and small subunits are transported to the cytoplasm

- (d)** Unlike eukaryotes, prokaryotes do not have endoplasmic reticulum or vesicles.

Suggest how prokaryotes are still able to secrete proteins. [1]

- Prokaryotes have protein channels in the cell surface membrane that allows secretion of proteins

[Total: 10]

QUESTION 3

Hereditary hypophosphatemic rickets is a genetic disorder that results on low level of phosphate in the blood (hypophosphatemia).

Fig. 3.1 shows the inheritance of this disease over four generations in an extended family.

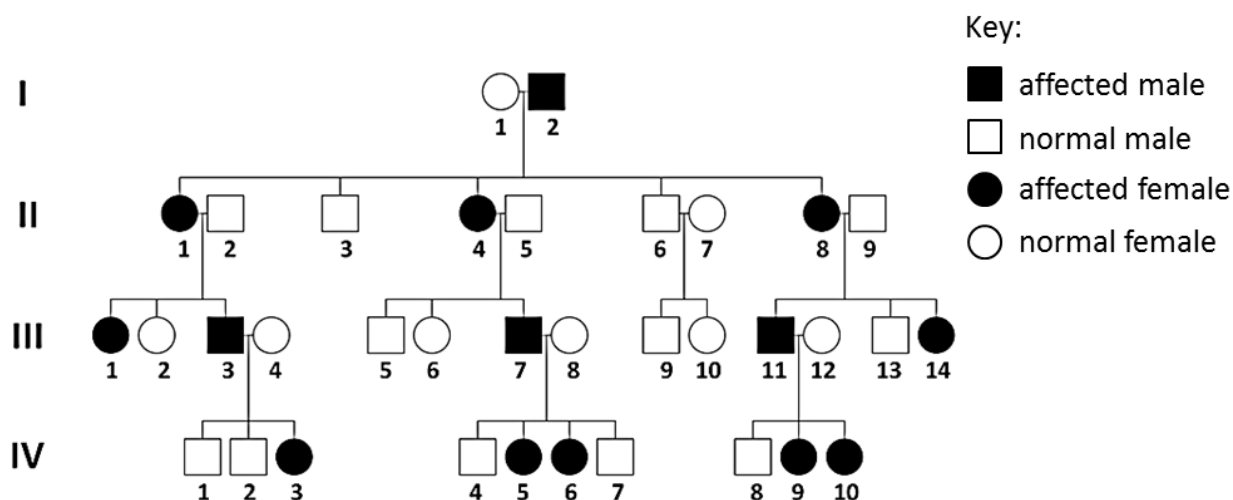


Fig. 3.1

(a) Explain the mode of inheritance of this disease.

[2]

- X-linked dominant / sex-linked dominant inheritance
- An affected male passes the disease allele on the X chromosome to all his daughters but not his son + **citing any one example**
 - o I-2 passed the dominant allele to daughters II-1, II-4, II-8 but not to the sons
 - o III-7 passed the dominant allele to daughters IV-5 and IV-6 but not to the sons
 - o III-3 passed the dominant allele to daughters IV-3 but not to the sons
 - o III-11 passed the dominant allele to daughters IV-9 and IV-10 but not to the sons

(b) Using the symbols **D/d** to represent the alleles, construct a genetic diagram to show how individual III-7 and III-8 can pass on the disease to their children.

[4]

Parental phenotype	Affected male		Unaffected female		
Parental genotype	$X^D Y$		$X^d X^d$		[1]
Gametes	X^D	Y	X^d	X^d	[1]
F1 genotype	$X^D X^d$	$X^D X^d$	$X^d Y$	$X^d Y$	
F1 phenotype	Affected female	Affected female	Unaffected male	Unaffected male	[1]
F1 phenotypic ratio	1 affected female : 1 unaffected male				[1]

Fig. 3.2 shows the distribution of malaria and the allele frequency of the sickle cell allele.

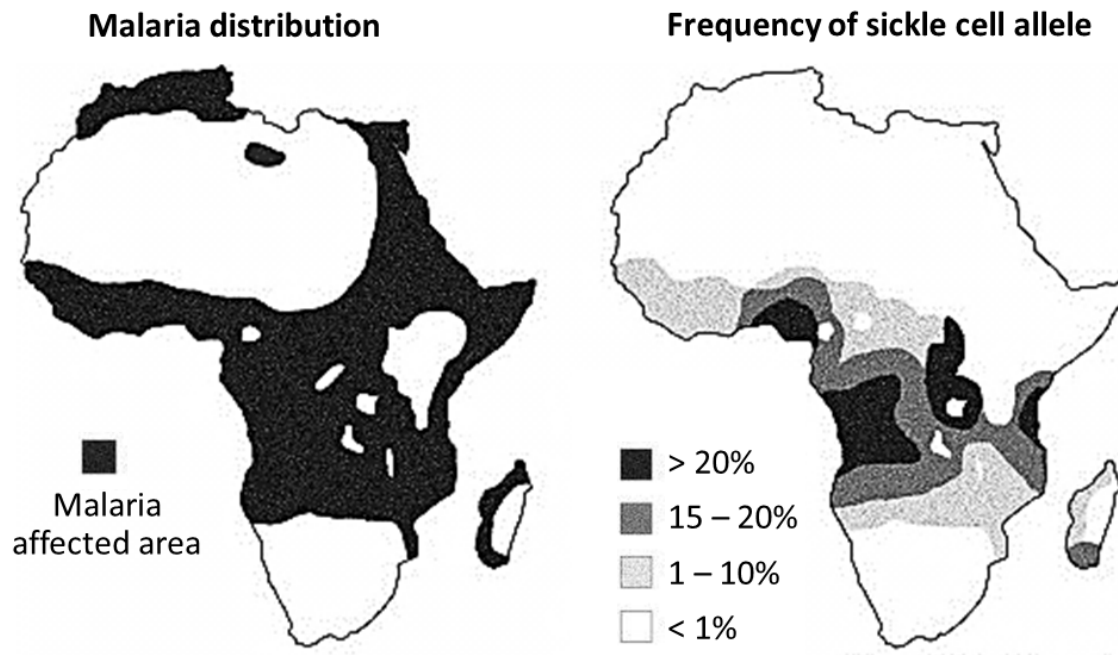


Fig. 3.2

(a) Describe and explain the correlation between the occurrence of malaria and the frequency of the sickle cell allele. [4]

- Frequency of sickle cell allele is higher in malaria affected areas
- Heterozygotes do not exhibit sickle cell anaemia / phenotypically normal
- Heterozygous / sickle cell traits individuals are at selective advantage in malaria affected areas / more resistant to malaria
- Heterozygous individuals survive to reproduce and pass on the sickle cell allele to viable and fertile offspring...
- ...hence increasing the HbS allele frequencies in the population within these malaria areas over time.

[Total: 10]

QUESTION 4

pCMV6-XL5 is a plasmid with a multiple cloning site (MCS) that lies downstream of the CMV promoter (Fig. 4.1). This plasmid can be inserted into both eukaryotic and prokaryotic host cells. The arrows denote the direction in which the genes are transcribed.

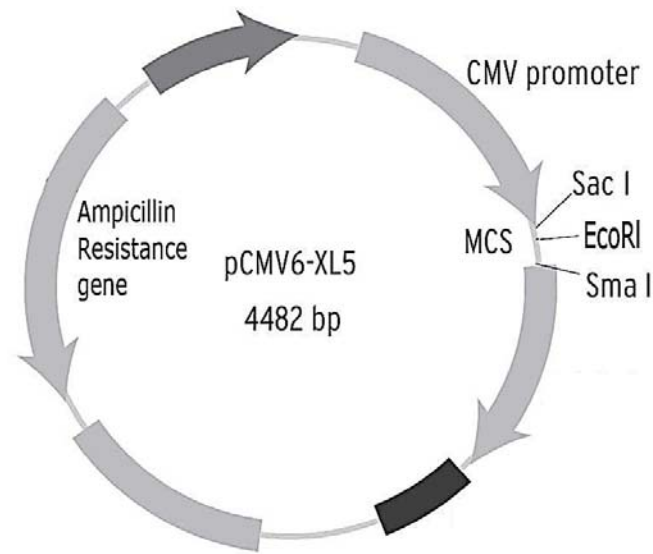


Fig. 4.1

(a) State the function of the CMV promoter in pCMV6-XL5.

[1]

- It is where **RNA polymerase binds** to **initiate transcription** of the human growth hormone gene inserted into the multiple cloning site (downstream of the promoter).

An artificially-synthesised human growth hormone (hGH) gene with flanking *Hind*III restriction site sequences was created. The restriction sites for the restriction enzymes *Hind*III and *Sac*I are shown in Fig. 4.2.

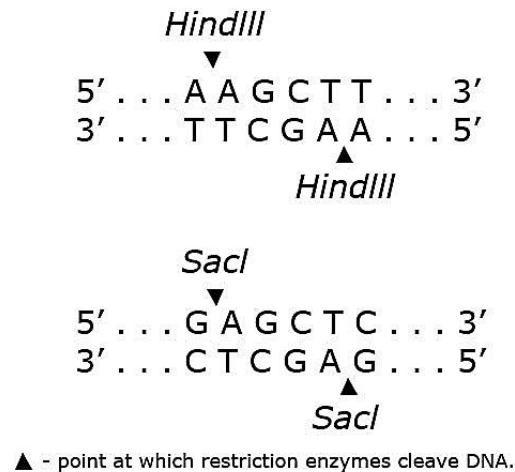


Fig. 4.2

(b) Using information in Fig. 4.1 and 4.2, explain how the hGH gene can be inserted into pCMV6-XL5. [3]

- Cleave gene using *Hind*III to generate sticky ends.
- Cleave plasmid with *Sac*I to generate complementary sticky ends to the *Hind*III sticky ends flanking the hGH gene.
- Mix the cleaved gene and plasmid together and add DNA ligase to seal the nicks / form phosphodiester bond between gene and plasmid.

(c) Describe the steps required to select bacterial colonies containing the human growth hormone gene after artificial transformation has taken place. [5]

1. Spread/plate transformed bacteria onto agar/nutrient medium plates containing ampicillin.
2. Bacterial colonies on the ampicillin plate are transferred onto a nitrocellulose membrane.
3. Add sodium hydroxide to the filter to lyse bacterial cells and denature dsDNA into ssDNA.
4. A solution of radioactively-labelled gene probe is added so that it will hybridize with the complementary DNA sequence of the human growth hormone gene.
5. Wash away excess probe
6. The position of the cell containing the DNA-gene probe hybrid/labelled gene can be located by autoradiography performed on the filter.
7. The position of dark spots on the x-ray film corresponds to the position of the bacterial colonies containing the human growth hormone gene.

Accept: Fluorescent-labelled gene probes and subsequent corresponding steps.

(d) Plasmid replication is usually initiated by RNAll, which is a sequence of ribonucleotides that binds to plasmid DNA prior to replication.

State and explain the role of RNAll in plasmid replication. [2]

- RNAll serves as a **primer** for DNA replication.
- Because **DNA polymerase** requires a **3'OH** to bind to in order to start/begin replication.

[Total: 11]

Section B
Answer **ONE** question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in questions **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 5

(a) Explain how genetically modified organisms may provide a solution to global food shortage issues. [8]

1. **Increase yield/harvest** of crops by genetically modifying food crops/inserting genes that will result in increased size/growth rate of food animals, or by decreasing losses caused by insect pests, herbicides, viruses, fungal infestation, and damage due to transportation of fully ripened fruits.
2. Genetically engineer Atlantic **salmon** by the insertion of a **growth hormone gene** from a Pacific Chinook salmon and an active promoter from an ocean pout placed upstream of the growth hormone gene.
3. The expression of the growth hormone gene results in **faster growth rate** and **yield** of the salmon, thereby increasing the supply of salmon.
4. Genetically engineer **maize/potato** by insertion of a **Bt-toxin gene** from the bacteria *Bacillus thuringiensis* that codes for the production of a crystalline Bt-toxin protein.
5. The **Bt-toxin protein** kills insect larvae feeding on the plant / allow plants to have **resistance** to a range of **insect** pests.
6. Genetically engineer **alfafa/maize** by insertion of a **herbicide resistance gene**/EPSPS enzyme gene.
7. The genetically engineered crop plant will be resistant to herbicide/glyphosate, such that when the herbicide/glyphosate is applied to the field, **only the weeds are eliminated**.
8. Genetically engineer **tobacco** by inserting a **tobacco mosaic virus (TMV) coat protein gene**. Transgenic plants that express the coat protein become resistant to infection by that virus.
Or
Genetically engineer **papaya** by inserting a **papaya ringspot virus (PRSV) coat protein gene**. Transgenic plants that express the coat protein become resistant to infection by that virus.
Or
Genetically engineer **plum** by inserting a **plum pox virus (PPV) coat protein gene**. Transgenic plants that express the coat protein become resistant to infection by that virus.
9. Genetically engineer **alfafa/potato** by inserting a **gene expressing anti-fungal compounds**/phytoalexin. Expression of the gene confers fungal resistance to the plant.

10. Genetically engineer **tomatoes** (Flavr Savr tomatoes) by inserting a **gene for antisense mRNA to polygalacturonase gene**.
11. The antisense mRNA reduces the expression of the polygalacturonase gene / complementary base pairs to the polygalacturonase mRNA thus the production of the polygalacturonase enzyme.
12. This will result in **delayed ripening** allowing crops to be stored for longer period of time/ less damage from transportation of fully ripened fruits.

(b) Explain how reduced NADP is formed in plant cell and how it is used in photosynthesis. [5]

1. Photolysis of water releases H^+ ion in the thylakoid space.
2. Special chlorophyll a from P680/ PSII and P700/PSI released electrons during photoactivation.
3. The electrons flow down the electron transport chain to combine with H^+ and $NADP^+$ from the stroma to form NADPH
4. The reaction is catalysed by NADP reductase.
5. Reduced NADP is used to reduce glycerate-3-phosphate to glyceraldehyde-3-phosphate in Calvin Cycle/light independent reaction
6. Reduced NADP is oxidised to regenerate NADP for light-dependent reaction

(c) Explain how molecular homology supports Darwin's theory of natural selection. [7]

1. Similarities in homologous molecules imply a common ancestry.
2. Phylogenetic relationships can be studied using the analysis of the sequence of amino acids in protein and nucleotide sequence in DNA/RNA of the organism
3. The molecular differences between homologous molecules of related species are due to different selection pressures in different environments acting on existing variation / alleles.
4. Over a long time/many generations, changes in allele frequencies occur/accumulation of advantageous allele in the population.
5. Mutations lead to changes in nucleotide sequences / changes in amino acid sequences in proteins, which is evident that evolution occurred.
6. The greater the difference in homology, the more distantly related are two species
7. E.g. Structure of alpha-globin, beta-globin in haemoglobin or cytochrome b/c

[Total: 20]

QUESTION 6

- (a) Describe and explain how the structures of chloroplasts and mitochondria are similarly adapted for their functions. [8]

	Structural similarity	Function
1	The <u>presence of electron carriers</u> on the <u>thylakoid membrane</u> and <u>mitochondrial inner membrane</u>	Passage of electrons via the electron carriers <u>release energy</u> to create proton gradient for the synthesis of ATP
2	The <u>presence of proton pump</u> on the thylakoid membrane and mitochondrial inner membrane	<u>Pump protons</u> from the stroma to thylakoid space in chloroplast and matrix to inter-membrane in mitochondria to create a <u>proton gradient</u>
3	The <u>inner membrane of mitochondria</u> and the <u>thylakoid membrane</u> is <u>extensively folded</u>	To <u>increase surface area</u> for attachment of e.g. electron carriers / photosynthetic pigments
4	Both contain <u>ATP synthase</u> on the thylakoid membrane and mitochondrial inner membrane	To allow the <u>facilitated diffusion of protons</u> to generate ATP To allow <u>phosphorylation of ADP to ATP</u>
5	Both contain <u>circular DNA</u>	To contain <u>genes</u> that code for proteins/enzymes involved in photosynthesis and respiration
6	Both contains <u>70S ribosomes</u>	For <u>translation</u> of mRNA into proteins/enzymes involved in photosynthesis and respiration
7	Both are <u>double-membrane</u> bound organelles	For <u>compartmentalization</u> and localization of enzymes for Krebs Cycle and link reaction in mitochondria matrix and Calvin Cycle in chloroplast stroma

(b) Compare the features of zygotic stem cells and adult stem cells.

[6]

Differences

Features	Zygotic	Adult
Source	Isolated from the 1-8 cell stage	Isolated from the adult organs/tissues
Potency	Totipotent	Multipotent
Differentiation	Capable of developing into all tissues, including those of the extra-embryonic membranes	Can give rise to a limited range of cells types.
Self-renewal capability	Undergoes self-renewal until the 8-cell stage.	Able to undergo self-renewal many more times.
Function	Differentiate into all types of cells to form the foetus.	For the growth and repair of tissues.

Similarities

1. Unspecialised cells
2. Self-renewal (mitosis) to give more stem cells
3. Able to give rise to fully differentiated cells.

(c) Discuss the goals of the Human Genome Project.

[6]

1. Identify all the genes in human DNA (approximately 20000 -25000).
2. Determine the DNA sequence of the entire human genome (about 3 billion base pairs).
3. Store this information in databases to be shared publicly by researchers throughout the world
4. Improve tools for data analysis.
5. Transfer related technologies to the private sector and award grants for innovative research, so as to catalyse the biotechnology industry and the development of new medical applications.
6. Address the ethical, legal and social issues that may arise from the project.

[Total: 20]